

Chapter 03 – Making Decisions: The Selection Structure

Full Exam Paper with Answers

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Part I – Multiple Choice Questions (20 Marks)

1. 1. Which symbol is used to define Template Literals in JavaScript?

- A) Single quotes (')
- B) Double quotes (")
- C) Backticks (` `)
- D) Parentheses ()

☒ Correct answer: C

2. 2. Template Literals were introduced in:

- A) ES5
- B) ES6
- C) ES7
- D) ES8

☒ Correct answer: B

3. 3. What is the main advantage of Template Literals?

- A) Data validation
- B) String interpolation
- C) Data encryption
- D) Error handling

☒ Correct answer: B

4. 4. Variable swapping can be done using:

- A) A temporary variable
- B) Arithmetic operations
- C) Both A and B
- D) None

☒ Correct answer: C

5. 5. Which keyword is used in a single-alternative selection?

- A) else
- B) switch
- C) if
- D) case

☒ Correct answer: C

6. 6. Dual-alternative selection uses:

- A) if
- B) else
- C) if...else
- D) switch

☒ Correct answer: C

7. 7. Multiple-alternative structure can be written using:

- A) switch
- B) nested if
- C) else if
- D) All of the above

☒ Correct answer: D

8. 8. What will the code output if temp = 25?

```
if(temp < 32){console.log('Snow');}else{console.log('No snow');}
```

- A) Snow
- B) No snow
- C) Error
- D) None

☒ Correct answer: B

9. 9. Which operator checks both value and type?

- A) ==
- B) =
- C) ===
- D) !==

☒ Correct answer: C

10. 10. The logical AND operator is represented by:

- A) ||
- B) &&
- C) !
- D) ==

☒ Correct answer: B

11. 11. The logical OR operator is represented by:

- A) ||
- B) &&
- C) !
- D) ==

☒ Correct answer: A

12. 12. The NOT operator reverses a condition and is written as:

- A) ||
- B) &&
- C) !
- D) ==

☒ Correct answer: C

13. 13. Which is the correct syntax for nested if?

- A) if(if()){}
- B) if() if(){}
- C) if(){ if(){} }
- D) nested if(){}

☒ Correct answer: C

14. 14. The ternary operator is represented as:

- A) if/else
- B) ? :
- C) switch
- D) ==

☒ Correct answer: B

15. 15. In a switch statement, each case should end with:

- A) end
- B) stop

- C) break
- D) default

☒ Correct answer: C

16. 16. What happens if break is omitted in a switch?

- A) Only one case runs
- B) All cases run until break or end
- C) Program stops
- D) None

☒ Correct answer: B

17. 17. Which structure is most suitable for many conditions?

- A) if
- B) if...else if
- C) switch
- D) ternary

☒ Correct answer: C

18. 18. The ternary operator is used to:

- A) Loop code
- B) Assign values based on condition
- C) Stop execution
- D) None

☒ Correct answer: B

19. 19. Which logical operator returns true only if both sides are true?

- A) ||
- B) &&
- C) !
- D) ==

☒ Correct answer: B

20. 20. In 'Overtime Pay' example, which condition checks both hours and rate?

- A) (hours > 40 && rate < 20)
- B) (hours < 40 || rate > 20)
- C) (hours >= 40 && rate > 20)
- D) (hours <= 40 || rate <= 20)

☒ Correct answer: A

Part II – True or False (20 Marks)

1. Template literals use backticks (`).

Answer: True

2. Template literals cannot contain variables.

Answer: False

3. Variable swapping can be done using arithmetic operations.

Answer: True

4. A single-alternative structure executes only when the condition is true.

Answer: True

5. Dual-alternative selection uses if and else.

Answer: True

6. Multiple-alternative selection allows only one if statement.

Answer: False

7. Logical AND is represented by &&.

Answer: True

8. Logical OR is represented by ||.

Answer: True

9. Logical NOT is represented by !.

Answer: True

10. The ternary operator uses ? and : symbols.

Answer: True

11. Switch structure replaces multiple if-else chains.

Answer: True

12. Omitting break in switch executes multiple cases.

Answer: True

13. The default keyword handles unmatched cases in switch.

Answer: True

14. Compound conditions combine multiple tests.

Answer: True

15. Overtime pay uses logical AND to check two conditions.

Answer: True

16. Nested if allows decisions inside another decision.

Answer: True

17. Template literals cannot span multiple lines.

Answer: False

18. if statement checks whether a condition is true.

Answer: True

19. else statement executes when condition is false.

Answer: True

20. switch statement must always include a default.

Answer: False

Part III – Direct Questions (20 Marks)

1. Define Template Literals.

Answer: Strings enclosed by backticks that allow embedding variables or expressions.

2. Mention two advantages of Template Literals.

Answer: String interpolation and multiline strings.

3. Explain variable swapping.

Answer: It is the process of exchanging values between two variables.

4. List two ways to perform variable swapping.

Answer: Using a temporary variable or arithmetic operations.

5. Define a conditional statement.

Answer: A statement that executes code based on a true or false test.

6. Write syntax for an if statement.

Answer: `if (condition) { code }`

7. Differentiate between single and dual alternative structures.

Answer: Single executes one block; dual executes one of two blocks.

8. Explain multiple-alternative structures.

Answer: They check several conditions sequentially using `if...else if`.

9. What is a compound condition?

Answer: Combining two or more conditions using logical operators.

10. List three logical operators.

Answer: `&&`, `||`, and `!`

11. What does the ternary operator do?

Answer: Assigns a value depending on whether a condition is true or false.

12. Write an example using the ternary operator.

Answer: `let msg = (x > 5) ? 'Yes' : 'No';`

13. Explain the function of a switch statement.

Answer: It executes different code blocks based on matching case values.

14. Why is the break statement important in a switch?

Answer: It prevents executing subsequent cases unintentionally.

15. What is the role of the default case?

Answer: It executes if no other case matches.

16. Describe nested if statements.

Answer: An if statement placed inside another if statement for multiple conditions.

17. Define compound conditions with example.

Answer: `if (price > 5 && price < 12)` checks both conditions together.

18. Explain the overtime pay example.

Answer: It uses logical AND to calculate extra pay when hours > 40 and rate < 20.

19. Write the syntax for switch statement.

Answer: `switch(expression) { case x: code; break; default: code; }`

20. Differentiate between if...else if and switch structures.

Answer: if...else if tests ranges or complex conditions; switch tests fixed values.